

A Piezoelectric-Driven High-Throughput Microfluidic Platform for Optimizing Biomanufacturing Process for Cell-based Immunotherapy Cancer Treatment

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Background

Clinical Unmet Needs:

- Chimeric Antigen Receptor (CAR) T-cell therapy is a promising immunotherapy for treating several types of cancer.¹
- Successful therapy depends on expanding large numbers of functional T-cells outside the body before reinfusion into patients.
- Current T-cell expansion methods require long culture times, significant resources, and high manufacturing costs.
- Improving T-cell proliferation is a key challenge for making CAR T-cell therapies more scalable and accessible.

LIV as a Potential Solution:

- Low-intensity vibration (LIV) is a non-drug mechanical stimulation approach that delivers controlled sinusoidal signals to biological systems.
- LIV signals are defined by parameters such as frequency, acceleration, displacement, treatment duration, and refractory period.
- By mimicking mechanical stimulation naturally produced during exercise, LIV may influence cellular activity and promote T-cell proliferation.
- Previous lab studies using voice-coil actuator LIV devices showed up to a 29% increase in T-cell proliferation compared to non-LIV controls.²

Needs for Optimization of LIV Signals:

- T-cell responses to LIV may vary based on cell type, cell size, patient age, immune status, health condition, and disease state, suggesting LIV is not a “one-size-fits-all” treatment.
- Identifying optimal LIV conditions requires testing multiple vibration parameters in parallel.
- Current voice-coil actuator systems are too large for high-throughput optimization and require large numbers of primary T-cells, which are difficult and expensive to obtain.

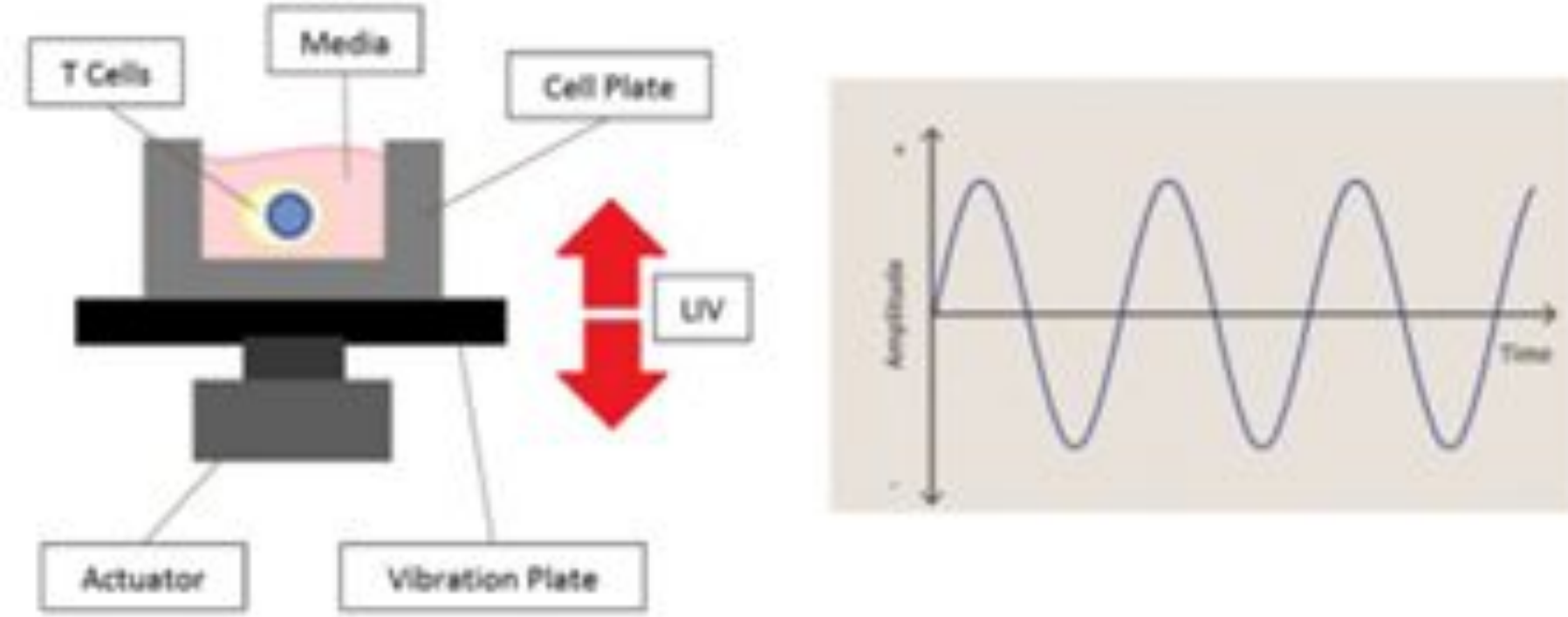


Fig 1. Typical LIV experimental setup and signal parameters.⁴

Design Objectives

- Develop PDMS-based microfluidic devices with multiple culture chambers to support parallel T-cell proliferation experiments using small cell populations.
- Adapt a proven primary T-cell microfluidic device that retains T-cells inside of the device during media renewal while limiting washout and shear stress.
- Integrate a piezoelectric actuator system to deliver controlled LIV across tunable parameters, including frequency, acceleration, displacement, treatment duration, and refractory period.
- Establish a PDMS casting workflow that allows microchannel and culture chamber geometry to be replicated and bonded to glass for live cell imaging.
- Establish a compact, high-throughput platform for optimizing LIV signal parameters while reducing device size and primary T-cell requirements.

Microfluidics Device Design

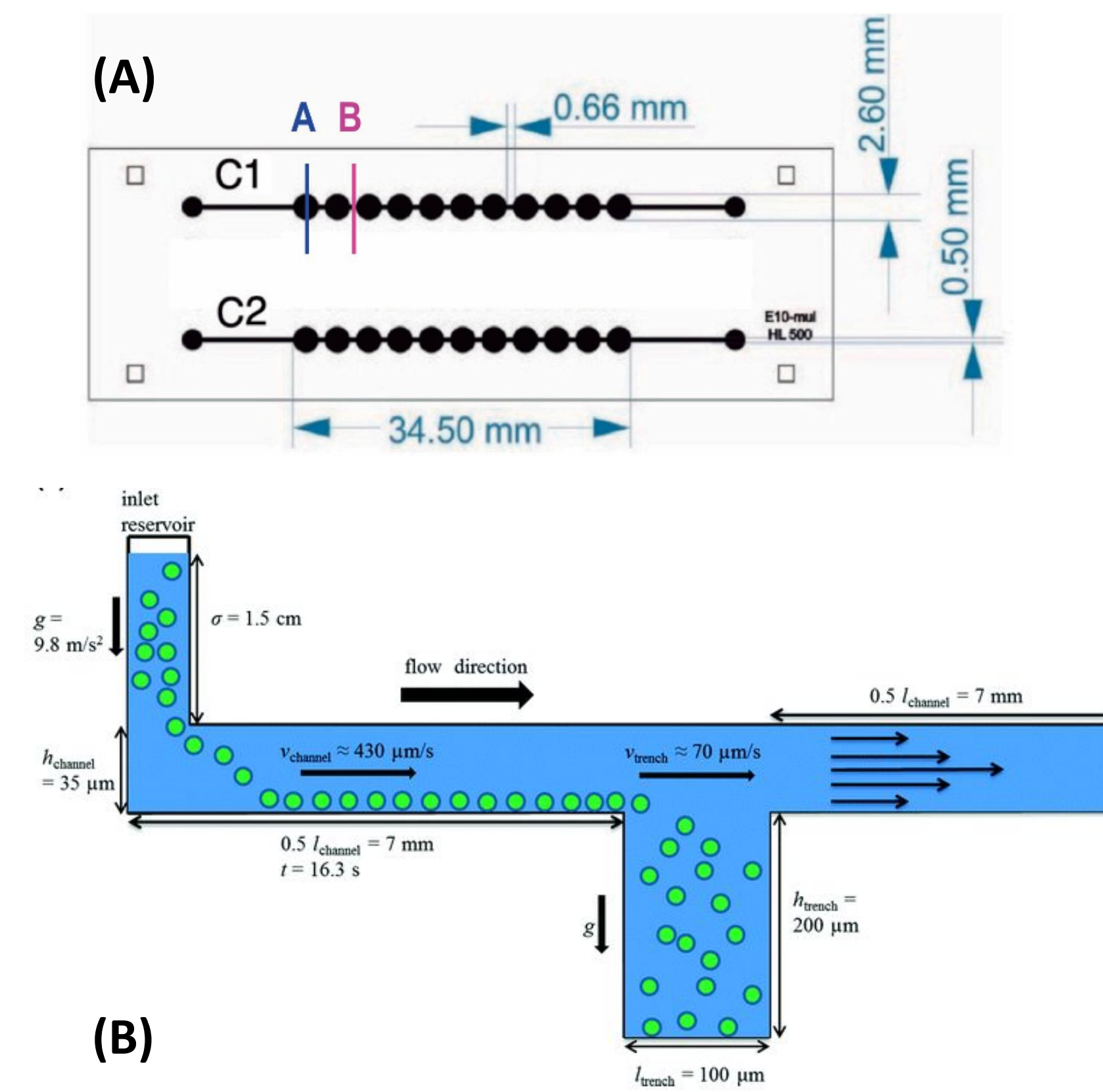


Fig. 2. Microfluidic device design and trapping mechanism. (A) Primary T-cell microfluidic culture device with parallel channels and repeated microchambers/wells for cell retention, media renewal, and long-term imaging.⁵ (B) Lab-in-a-Trench concept showing gravity-based cell settling into trench regions during media flow.³

- Device design is adapted from a multilevel microfluidic platform shown to successfully culture and monitor primary human T-cells over multiple days.⁵
- The chip uses parallel microchannels with small microchambers/wells that physically trap suspension T-cells and prevent them from being washed away during media renewal.
- Cells settle into the deeper center of each well, where they are protected from flow-induced shear stress while still receiving fresh culture media.
- This design enables long-term imaging of the same T-cell clusters, allowing cell growth and proliferation to be tracked over time.

Microfluidics Device Fabrication

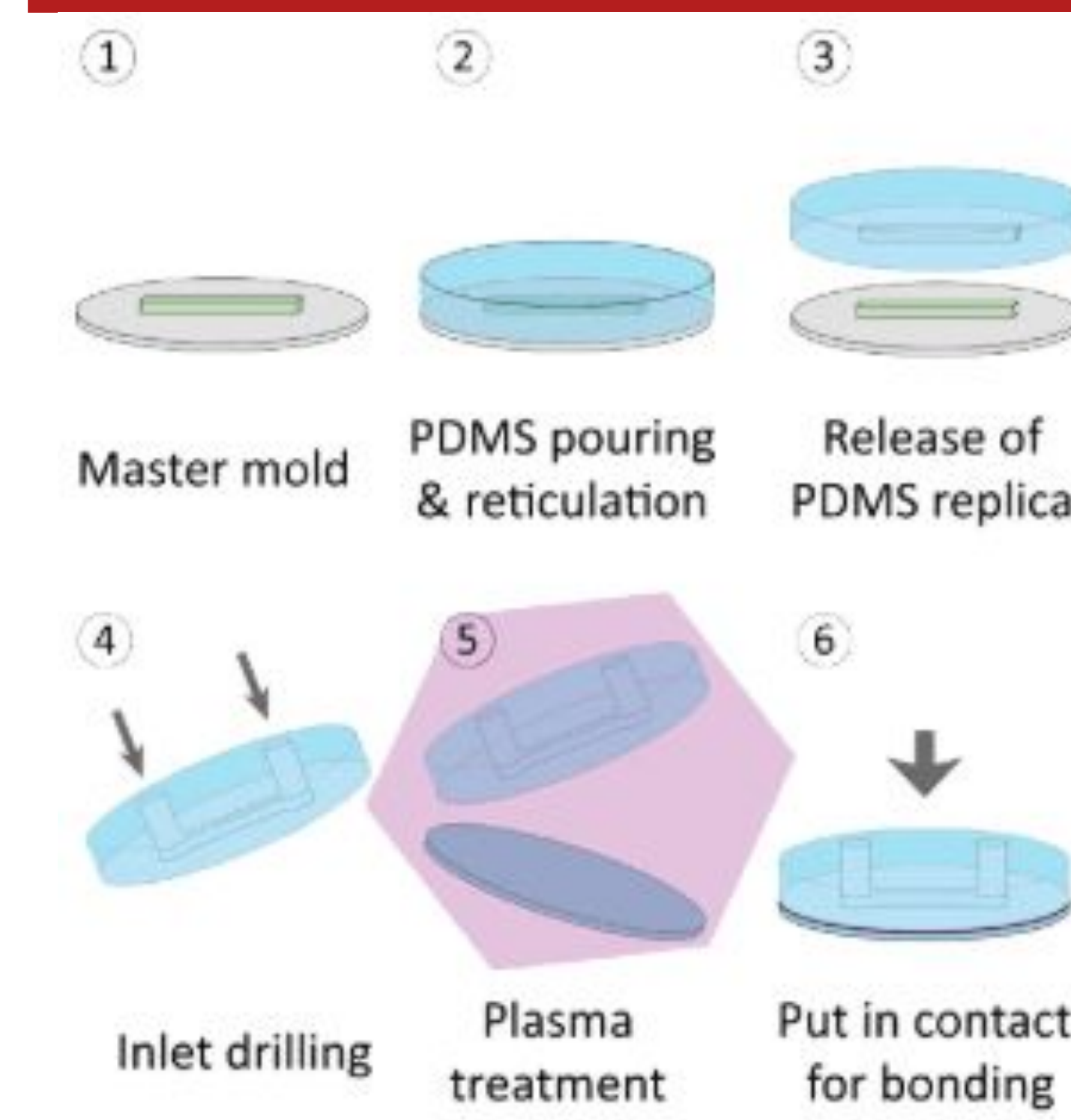


Fig. 3. Proposed PDMS casting workflow.⁶ PDMS is poured over a master mold, cured to replicate the microchannel features, released from the mold, punched to create inlet/outlet ports, plasma-treated, and bonded to glass to form a sealed microfluidic device.

- Microfluidic device fabrication will follow the PDMS casting workflow.
- A master mold will be used to replicate the microchannel and culture chamber geometry.
- The master mold fabrication method is still being evaluated, with options including SU-8 lithography, 3D printing, Two-Photon Polymerization, or epoxy replica molding.
- The final PDMS device will be plasma-bonded to glass and integrated with the piezoelectric LIV platform.

Piezoelectric Actuator

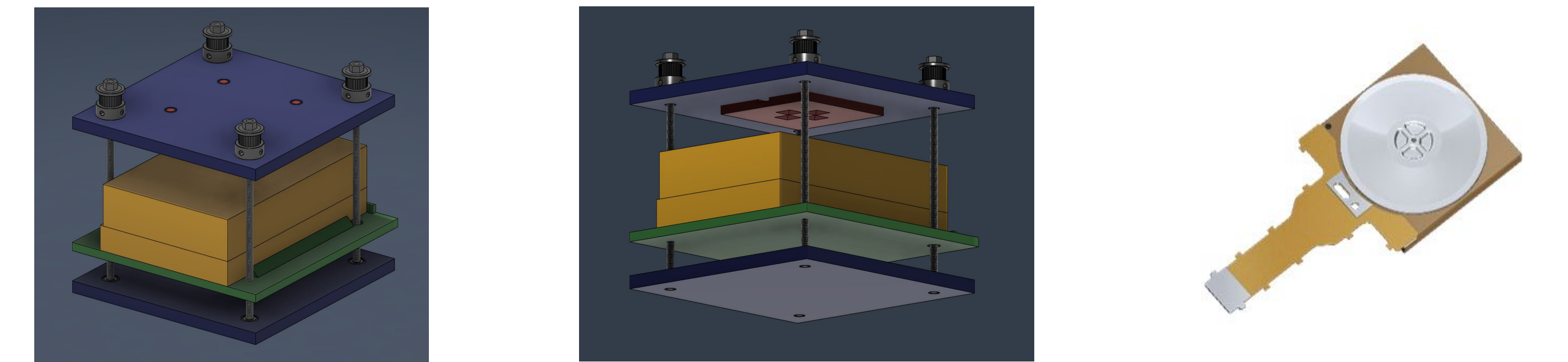


Fig 4. 3D model of the proposed piezoelectric actuator platform and the selected actuator for LIV stimulation.

- Custom 3D-printed platform designed to support the microfluidic device during LIV stimulation.
- Four corner lead screws with hex nut inserts provide alignment, stability, and adjustable positioning.
- A springboard supports the device and allows controlled displacement.
- A piezoelectric actuator makes direct contact with the microfluidic device to deliver vibration.

Future Work/Long-term Goals

- Select and finalize the master mold fabrication method for PDMS device casting.
- Fabricate and test PDMS microfluidic devices for channel integrity, bonding quality, and media flow.
- Calibrate the piezoelectric platform to verify LIV parameters, including frequency, acceleration, and displacement.
- Optimize primary T-cell culture conditions based on the original microfluidic T-cell culture protocol and make necessary modifications for LIV stimulation experiments.
- Conduct control and LIV-treated T-cell culture experiments to evaluate growth, proliferation, and viability.
- Assess cell response using live-cell microscopy, CFSE proliferation assay, and Propidium Iodide (PI) viability assay.

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